

Homework 3 - CSCI633

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03/30/2026

Question 1. Clearly describe the intuition/motivation and basic operation behind each of the following algorithms:

1) Particle Swarm Optimization (PSO)

Intuition: PSO is inspired by the social behavior of birds flocking or schools of fishes. Individuals (particles) adjust their positions based on their own best known location (personal best) and the best location found by the entire swarm (global best). This mimics how birds share information to locate food sources.

Basic Operation: A swarm of N particles is initialized with random positions $\mathbf{x}_i$ and velocities $\mathbf{v}_i$. At each iteration, the velocity and position of each particle are updated as:

$$\mathbf{v}_i(t+1) = w\mathbf{v}_i(t) + c_1r_1(\mathbf{p}_i - \mathbf{x}_i(t)) + c_2r_2(\mathbf{g} - \mathbf{x}_i(t)),$$

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + \mathbf{v}_i(t+1).$$

- $\mathbf{p}_i$ is the particle's personal best position,
- $\mathbf{g}$ is the global best position among all particles,
- w is the inertia weight,
- c_1, c_2 are acceleration coefficients,
- r_1, r_2 are random numbers in $[0, 1]$.

2) Firefly Algorithm (FA)

Intuition: FA is based on the flashing behavior of fireflies. The light intensity is associated with the quality of the solution, brighter fireflies attract others. The attractiveness decreases with distance, leading to a natural balance between local exploitation and global exploration.

Basic Operation:

- Initialize a population of fireflies with random positions.
- Each firefly's brightness is determined by its objective function value.
- For each firefly i , compare with all brighter fireflies j . Move i toward j using:

$$\mathbf{x}_i = \mathbf{x}_i + \beta_0 e^{-\gamma r_{ij}^2} (\mathbf{x}_j - \mathbf{x}_i) + \alpha \cdot \mathbf{rand}$$

where r_{ij} is the distance, β_0 is initial attractiveness, γ controls attraction range, and α is a randomization parameter.

- If no brighter firefly exists, move randomly.
- Update brightness and repeat until stopping condition.

3) Differential Evolution (DE)

Intuition: DE is a population-based algorithm that uses differences between randomly selected individuals to create new candidates. It is simple and effective for continuous optimization problems.

Basic Operation:

- Initialize a population of random vectors.
- For each target vector $\mathbf{x}_i$, pick three distinct random indices r_1, r_2, r_3 .
- Create a mutant vector:

$$\mathbf{v} = \mathbf{x}_{r_1} + F(\mathbf{x}_{r_2} - \mathbf{x}_{r_3})$$

where F is the scaling factor.

- Form a trial vector $\mathbf{u}$ by binomial crossover: each component is taken from the mutant with probability Cr , otherwise from the target.
- If $\mathbf{u}$ has better fitness, it replaces $\mathbf{x}_i$.
- Repeat until stopping condition.

Question 2.

What are the primary advantages of the FA that result in its improved efficiency over other metaheuristics?

Simplicity and Ease of Implementation: The algorithm has only a few parameters $(\alpha, \beta_0, \gamma)$ and a straightforward pseudocode, making it accessible for a wide range of applications.

Distance-Dependent Attractiveness: The attractiveness decreases with distance which creates a natural balance. Nearby fireflies exert strong attraction (local exploitation), while far away fireflies have no influence. This allows the swarm to automatically divide into groups that can explore different optima which prevents premature convergence.

Generality and Flexibility. FA can be viewed as a generalization of several other metaheuristics. When $\gamma \rightarrow 0$ and the movement is directed toward the global best, it reduces to accelerated particle swarm optimization. When $\gamma \rightarrow \infty$ (attractiveness negligible), it behaves like simulated annealing. When $\gamma = 0$ and $\alpha = 0$, the update resembles differential evolution (DE). This flexibility allows FA to adapt to different problems.

What applications might these strongly benefit from using FA optimization?

Image Segmentation with Multilevel Thresholding: In image processing, segmentation often requires selecting optimal thresholds to separate objects from the background. For multilevel thresholding, the objective function can have many local optima. FA has the ability to automatically divide the population and explore multiple peaks simultaneously. It would be efficient at finding the best threshold combination for the image.

Job Shop Scheduling Problem (JSSP): In manufacturing, JSSP requires sequencing a set of jobs on a set of machines to minimize makespan. JSSP is NP-hard, with an enormous search space and many local optima. For this problem, FA is good because its distance-dependent attraction and automatic division allow multiple populations to be explored at once. The brightness of a firefly would correspond with the quality of a schedule. Attraction causes schedules that share beneficial orderings to move toward better schedules, while randomization prevents the whole swarm from converging prematurely.

Question 3. Implement differential evolution.

(a) Rosenbrock's Function

Population Size	F	Cr	$f(\hat{x})$ Mean $\pm$ Std
30	0.8	0.5	0.00000000 $\pm$ 0.00000000
30	1.0	0.3	0.00000000 $\pm$ 0.00000000
50	0.5	0.5	0.00000000 $\pm$ 0.00000000
50	0.8	0.3	0.00000000 $\pm$ 0.00000000

Discussion: For all parameter combinations tested, Differential Evolution consistently converged to the global optimum of Rosenbrock's function. The zero standard deviation indicates that every trial succeeded, the 2-D Rosenbrock problem is relatively easy for DE.

(b) Easom Function

Population Size	F	Cr	$f(\hat{x})$ Mean $\pm$ Std
30	0.8	0.5	-1.00000000 $\pm$ 0.00000000
30	1.0	0.3	-0.96666705 $\pm$ 0.18255894
30	1.0	0.9	-0.86952416 $\pm$ 0.32341426
50	0.5	0.5	-1.00000000 $\pm$ 0.00000000
50	1.0	0.5	-0.89618904 $\pm$ 0.30414866

Discussion: For Easom's function, DE achieved perfect convergence for most parameter settings, especially when $F \leq 0.8$. However, $F = 1.0$ led to overshoot, resulting in mean fitness values as high as -0.87 with large standard deviations. The crossover probability Cr had a minor effect, but higher Cr combined with $F = 1.0$ worsened the results. This shows that DE's sensitivity to the scaling factor when the objective has a very narrow optimum.

Question 4. Implement particle swarm optimization.

(a) Rosenbrock's Function

Swarm Size	Inertia w	$f(\hat{x})$ Mean $\pm$ Std
30	0.4	0.00000000 $\pm$ 0.00000000
30	0.7	0.00000000 $\pm$ 0.00000000
50	0.4	0.00000000 $\pm$ 0.00000000
50	0.7	0.00000000 $\pm$ 0.00000000
50	0.9	0.00000001 $\pm$ 0.00000001
100	0.4	0.00000000 $\pm$ 0.00000000
100	0.7	0.00000000 $\pm$ 0.00000000
100	0.9	0.00000000 $\pm$ 0.00000000

Discussion: For all tested swarm sizes and inertia weights, PSO consistently found the global optimum of Rosenbrock's function. The only slight deviation is negligible, so this shows that PSO is also very effective on this 2D benchmark.

(b) Easom Function

Swarm Size	Inertia w	$f(\hat{x})$ Mean $\pm$ Std
30	0.4	-1.00000000 $\pm$ 0.00000000
30	0.7	-1.00000000 $\pm$ 0.00000000
50	0.4	-1.00000000 $\pm$ 0.00000000
50	0.7	-1.00000000 $\pm$ 0.00000000
50	0.9	-1.00000000 $\pm$ 0.00000000
100	0.4	-1.00000000 $\pm$ 0.00000000
100	0.7	-1.00000000 $\pm$ 0.00000000
100	0.9	-1.00000000 $\pm$ 0.00000000

Discussion: For all tested swarm sizes and inertia weights, PSO reliably converged to the global optimum of Easom's function. The zero standard deviation confirms that every trial succeeded, meaning that PSO is effective on this problem.